Po-Han Chen

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Education

Stanford University — **Stanford, CA**
Ph.D. in Electrical Engineering — 2021–2026
Advisor: Priyanka Raina
Thesis Title: *Designing CGRAs for Sparse ML Acceleration and Scalable Physical Implementation*

National Tsing Hua University (NTHU) — **Hsinchu, Taiwan**
M.S. of Electrical Engineering — 2016–2018
B.S. of Electrical Engineering and Computer Science — 2012–2016

Chip Research Experiences

Zircon: Heterogeneous Accelerator SoC — Stanford, CA

• Served as SoC integrator and physical designer for a heterogeneous accelerator combining a specialized systolic array and a flexible coarse-grained reconfigurable array (CGRA)
• Integrated an ARM CPU, a global buffer, a 64×32 systolic array, and a 16×16 CGRA into a complete SoC
• Implemented the chip in Intel 16 nm technology and completed physical design, timing closure, and signoff

Opal: CGRA SoC for Sparse Machine Learning — Stanford, CA

• Led the tapeout of an Intel 16 nm CGRA SoC with primitive-level optimizations for sparse ML workloads
• Developed CGRA architectural optimizations that reduced runtime and energy by 66–79% across evaluated sparse-ML benchmarks
• Established and pipe-cleaned the first complete Intel 16 nm implementation flow in the group, spanning synthesis, physical design, timing signoff, DRC, and LVS
• Designed a custom PCB and performed post-silicon bring-up, functional validation, performance characterization, and power measurement
• Automated CGRA firmware generation and testing to enable rapid application deployment
• Presented the work at the CICC conference and the ISSCC poster sessions

Open-Source CGRA — Stanford, CA

• Taped out an 8×4 CGRA using the open-source SkyWater 130 nm process and the OpenROAD EDA tools
• Integrated OpenRAM SRAM macros and a provided harness SoC with our tool-generated CGRA array
• Demonstrated technology portability of our CGRA-generation tools across an open-source CMOS flow
• Presented the work at the ISCAS conference

SAPIENS: RRAM-Based Non-Volatile Associative Memory — Stanford, CA

• Built an FPGA-based validation platform to generate configurable test stimuli, control the chip, and verify measured outputs
• Developed an on-the-fly tuning and testing flow that enabled rapid post-silicon characterization
• Results published at the Symposium on VLSI Technology conference

Digital Refocusing Accelerator — NTHU, Hsinchu

• Developed a TSMC 40 nm ASIC for accelerating block-based digital light-field refocusing
• Designed a highly parallel microarchitecture for view synthesis, exploiting pixel-, line-, and view-level parallelism to achieve up to 350× speedup over CPU execution and enable real-time refocusing
• Contributed across RTL design, functional verification, logic synthesis, and post-silicon chip measurement
• Presented the work at the Symposium on VLSI Circuits conference

Work Experiences

SEG Pixel IP Design Team, Apple **Cupertino, CA**

Hardware Designer (Intern) 2022

• Explored and implemented hardware architectures for a newly developed internal algorithm, achieving significant area reduction compared with the legacy design.

Camera ISP Pipeline Hardware Design Team, MediaTek **Hsinchu, Taiwan**

Hardware Designer (Full Time) 2019–2020

• Designed and verified digital IPs for camera image signal processing (ISP) pipelines, including functional verification, constrained random tests, coverage analysis, and synthesis.

• Developed automation tools to streamline internal hardware design and verification workflows.

Selected Publications

Full list: https://pohantw.github.io

P.-H. Chen, A. Carsello, M. Horowitz, and P. Raina, "SPEC: A Scalable and Physical-Design-Friendly Architecture for CGRAs," *IEEE Open Journal of the Solid-State Circuits Society* **(OJ-SSCS)**, 2026.

P.-H. Chen, B. W. Cheng, M. Oduoza, Z. Xie, R. Lu, S. G. Ravipati, K. Koul, A. Carsello, Y. Mei, M. Horowitz, and P. Raina, "Opal: A 16-nm Coarse-Grained Reconfigurable Array SoC for Full Sparse Machine Learning Applications," *IEEE Solid-State Circuits Letters* **(SSC-L)**, 2025.

P.-H. Chen, B. W. Cheng, M. Oduoza, Z. Xie, K. Koul, S. G. Ravipati, Y. Mei, R. Lu, A. Carsello, M. Horowitz, and P. Raina, "Opal: A 16nm Coarse-Grained Reconfigurable Array for Full Sparse ML Applications," *IEEE Custom Integrated Circuits Conference* **(CICC)**, 2025.

P.-H. Chen, C. Tsao, and P. Raina, "An Open-Source 4×8 Coarse-Grained Reconfigurable Array Using SkyWater 130 nm Technology and Agile Hardware Design Flow," *IEEE International Symposium on Circuits and Systems* **(ISCAS)**, 2023.

P.-H. Chen, S.-W. Yang, and C.-T. Huang, "A 250-mW 5.4G-Rendered-Pixel/s Realistic Refocusing Processor for High-Performance Five-Camera Mobile Devices," *IEEE Open Journal of the Solid-State Circuits Society* **(OJ-SSCS)**, 2023.

P.-H. Chen, S.-W. Yang, S.-Y. Huang, L.-D. Chen, and C.-T. Huang, "A 250mW 5.4G-Novel-Pixel/s Photorealistic Refocusing Processor for Full-HD Five-Camera Applications," *IEEE Symposium on VLSI Circuits* **(VLSI)**, 2019.

• A. Carsello, J. Thomas, A. Nayak, **P.-H. Chen**, M. Horowitz, P. Raina, and C. Torng, "mflowgen: A Modular Flow Generator and Ecosystem for Community-Driven Physical Design", in *ACM/IEEE Design Automation Conference* **(DAC)**, 2022.

• H. Li, W.-C. Chen, A. Levy, C.-H. Wang, H. Wang, **P.-H. Chen**, W. Wan, et al., "SAPIENS: A 64-kb RRAM-Based Non-Volatile Associative Memory for One-Shot Learning and Inference at the Edge", in *IEEE Transactions on Electron Devices*, vol. 68, no. 12, pp. 6637–6643, 2021.

• H. Li, W.-C. Chen, A. Levy, C.-H. Wang, H. Wang, **P.-H. Chen**, W. Wan, et al., "One-Shot Learning with Memory-Augmented Neural Networks Using a 64-kbit, 118 GOPS/W RRAM-Based Non-Volatile Associative Memory", in *Symposium on VLSI Technology*, 2021.

Skills

HW/SW Languages: Verilog, SystemVerilog, C, Python, Tcl

EDA Tools & Others: Simulators (VCS, Xcelium), Linting (Spyglass, nLint), Synthesis (DC, Genus), P&R (Innovus, ICC), LEC, DRC/LVS, Verdi, Git